

Output :

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL
big@hell-na:~/Kishor/python_ai$ python3 lab7.py
Step-by-step jug contents to goal state:
Jug A: 0L, Jug B: 0L
Jug A: 0L, Jug B: 3L
Jug A: 3L, Jug B: 0L
Jug A: 3L, Jug B: 3L
Jug A: 5L, Jug B: 1L
Jug A: 0L, Jug B: 1L
Jug A: 1L, Jug B: 0L
big@hell-na:~/Kishor/python_ai$
```

Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  
big@hell-na:~/Kishor/python_ai$ python3 lab6.py  
The optimal value is: 3  
big@hell-na:~/Kishor/python_ai$
```

Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

big@hell-na:~/Kishor/python_ai$ python3 lab5.py
Graph adjacency list with costs:
A -> [('B', 1), ('C', 4), ('D', 3)]
B -> [('E', 2)]
C -> [('E', 1), ('F', 5)]
D -> [('F', 6)]
E -> [('F', 1)]
F -> []
Shortest path from start to goal using A* Search:
A -> C -> E -> F
big@hell-na:~/Kishor/python_ai$
```

Output :

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL

big@hell-na:~/Kishor/python_ai$ python3 lab4.py
Adjacency List:
A -> ['B', 'C']
B -> ['D', 'E']
C -> ['F']
D -> []
E -> []
F -> []
Enter start node: A
Enter destination node: F
Path from A to F:
A -> B -> E -> C -> F
```